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(54) **METHOD AND APPARATUS FOR  
REMOVING PARTICLES OR PHOTORESIST  
ON SUBSTRATES**

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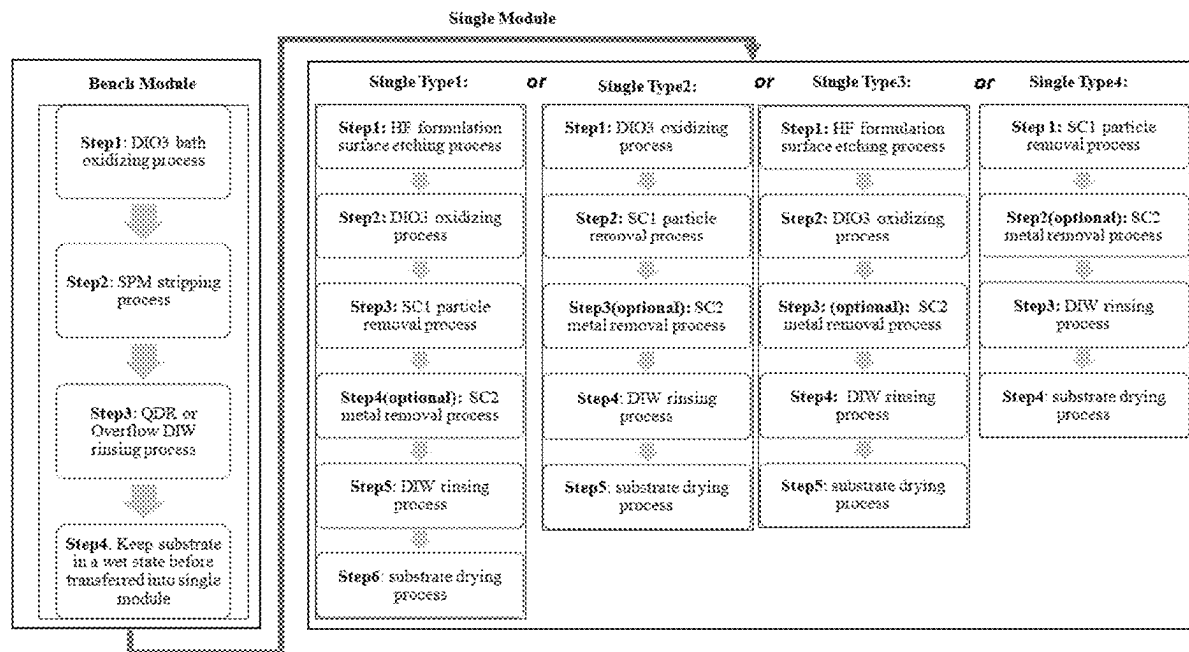
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21, 2022, filed as application No. PCT/CN2020/  
085948 on Apr. 21, 2020, now Pat. No. 12,334,367.

(57) **ABSTRACT**

An apparatus removes particles or photoresist on substrates. The apparatus includes a bench module, configured to implement one or more substrates bench cleaning process; a single module, having multiple single chambers configured to implement single substrate cleaning and drying process; and a process robot, configured to transfer the one or more substrates between the bench module and the single module. The bench module has at least one DIO<sub>3</sub> bath configured to accommodate DIO<sub>3</sub> solution for processing the one or more substrates, at least one SPM bath configured to accommodate SPM solution for processing the one or more substrates, at least one DIW bath configured to accommodate DIW for rinsing the one or more substrates, and at least one second substrate transfer robot configured to transfer the one or more substrates among the DIO<sub>3</sub> bath, the SPM bath and the DIW bath.



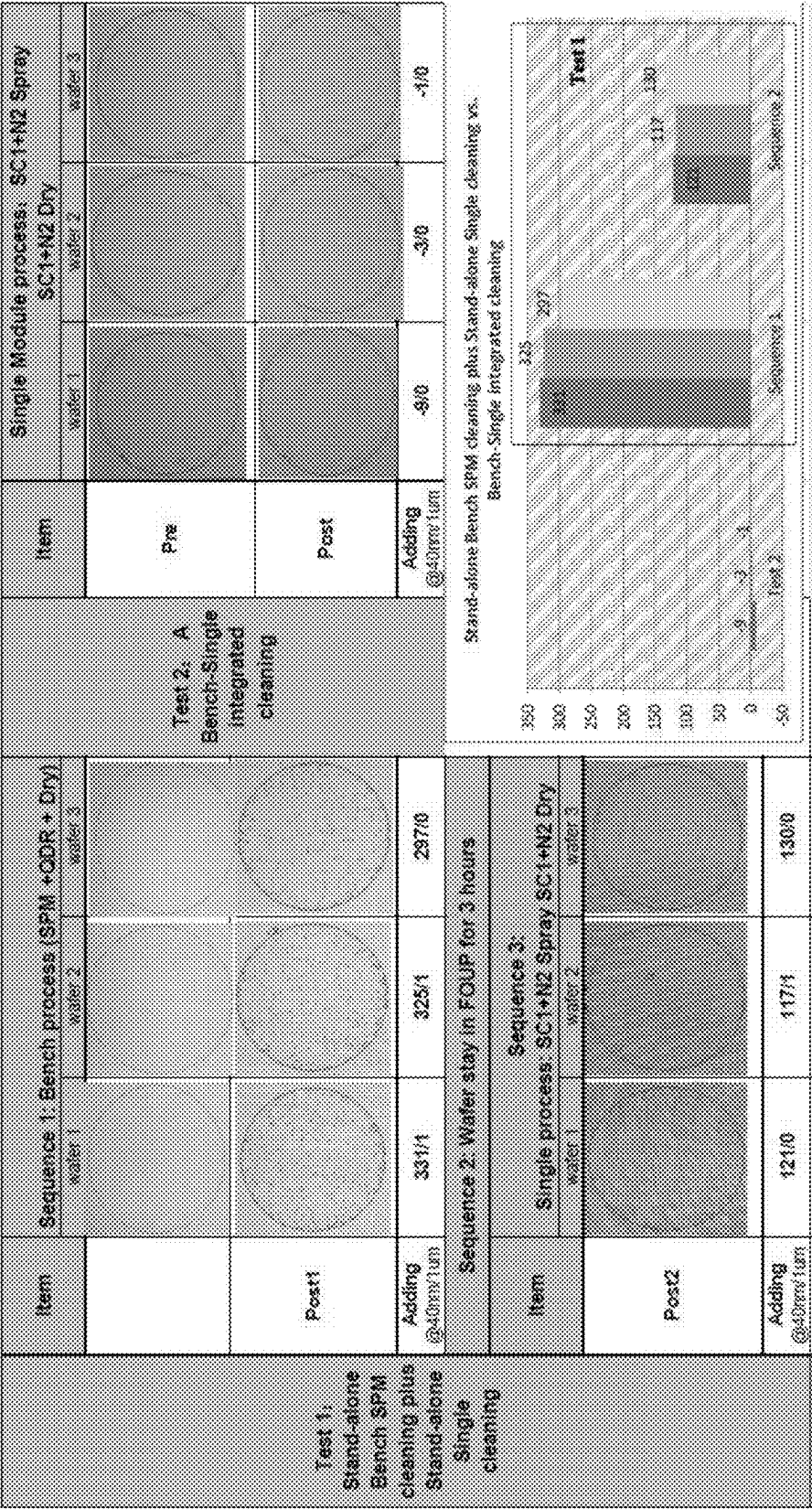


FIG 1

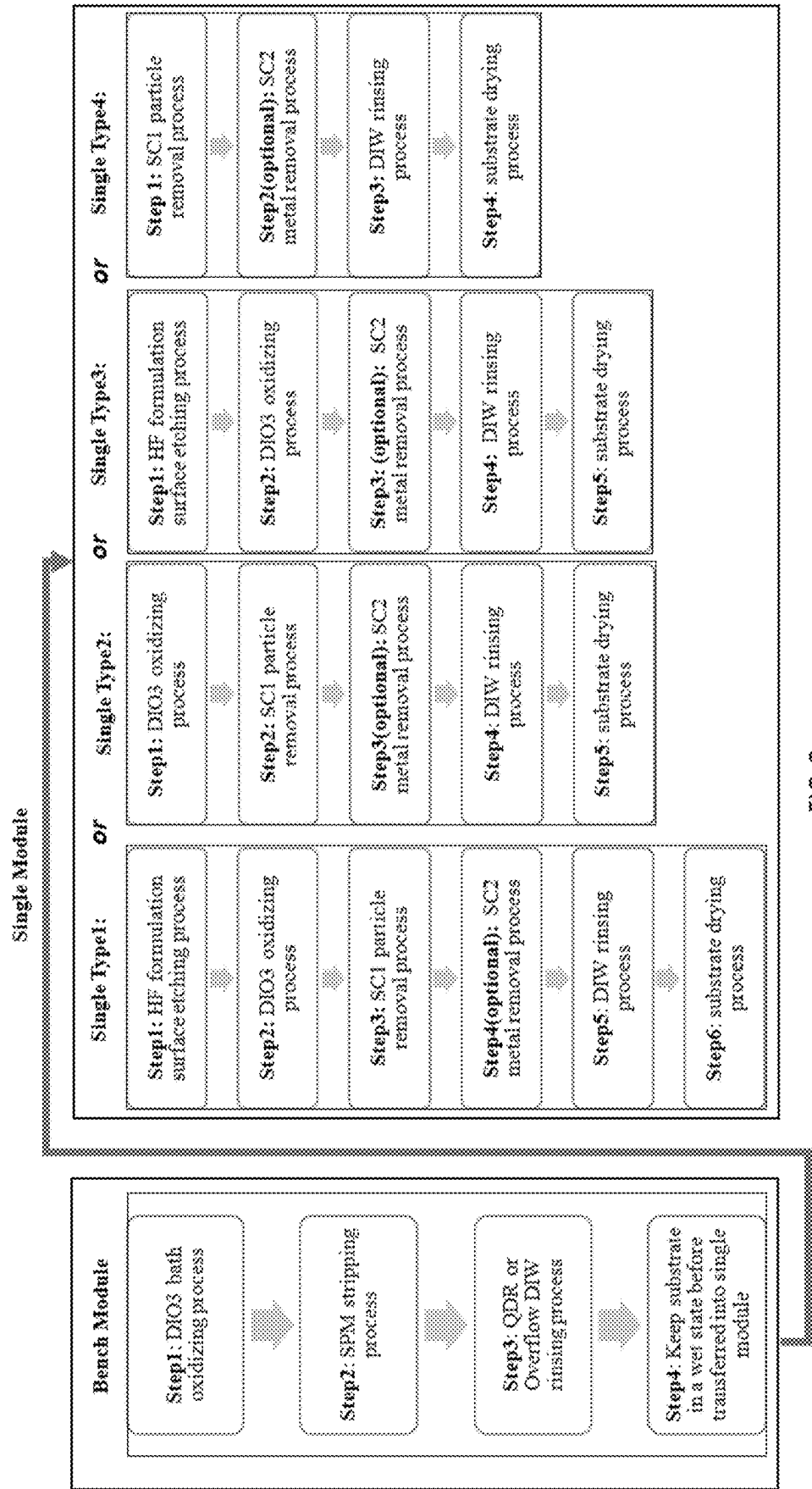
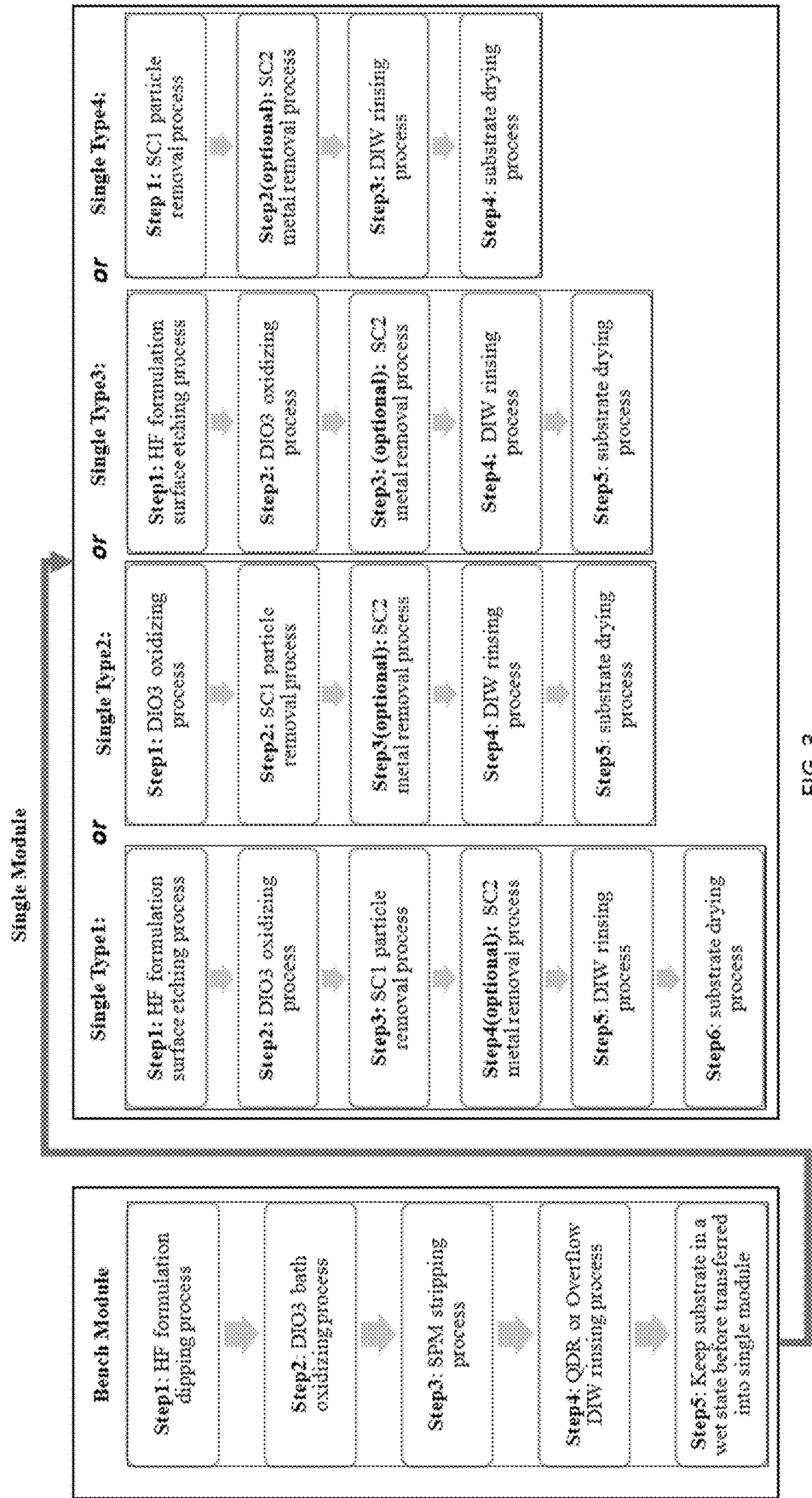


FIG 2



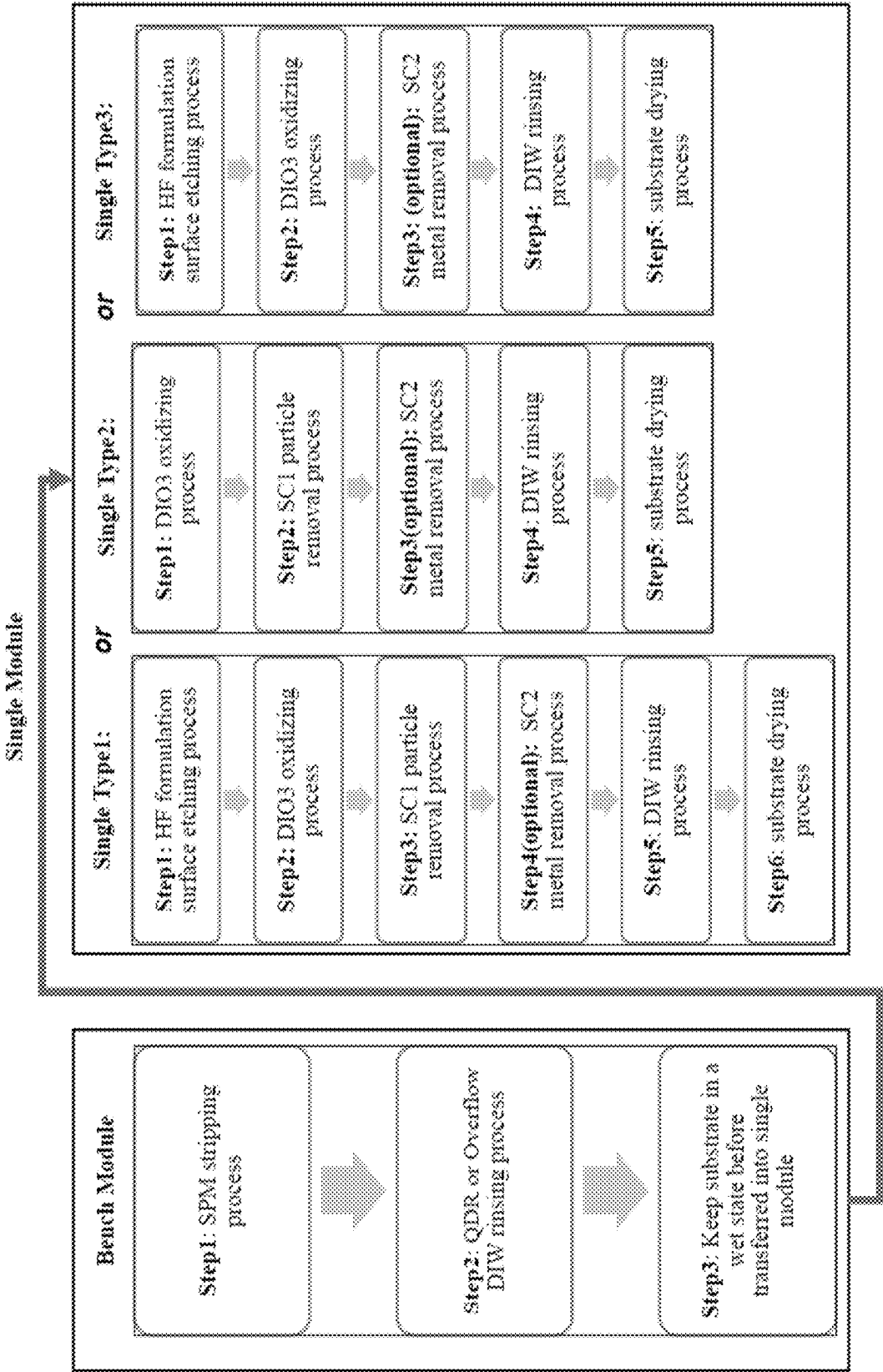


FIG 4

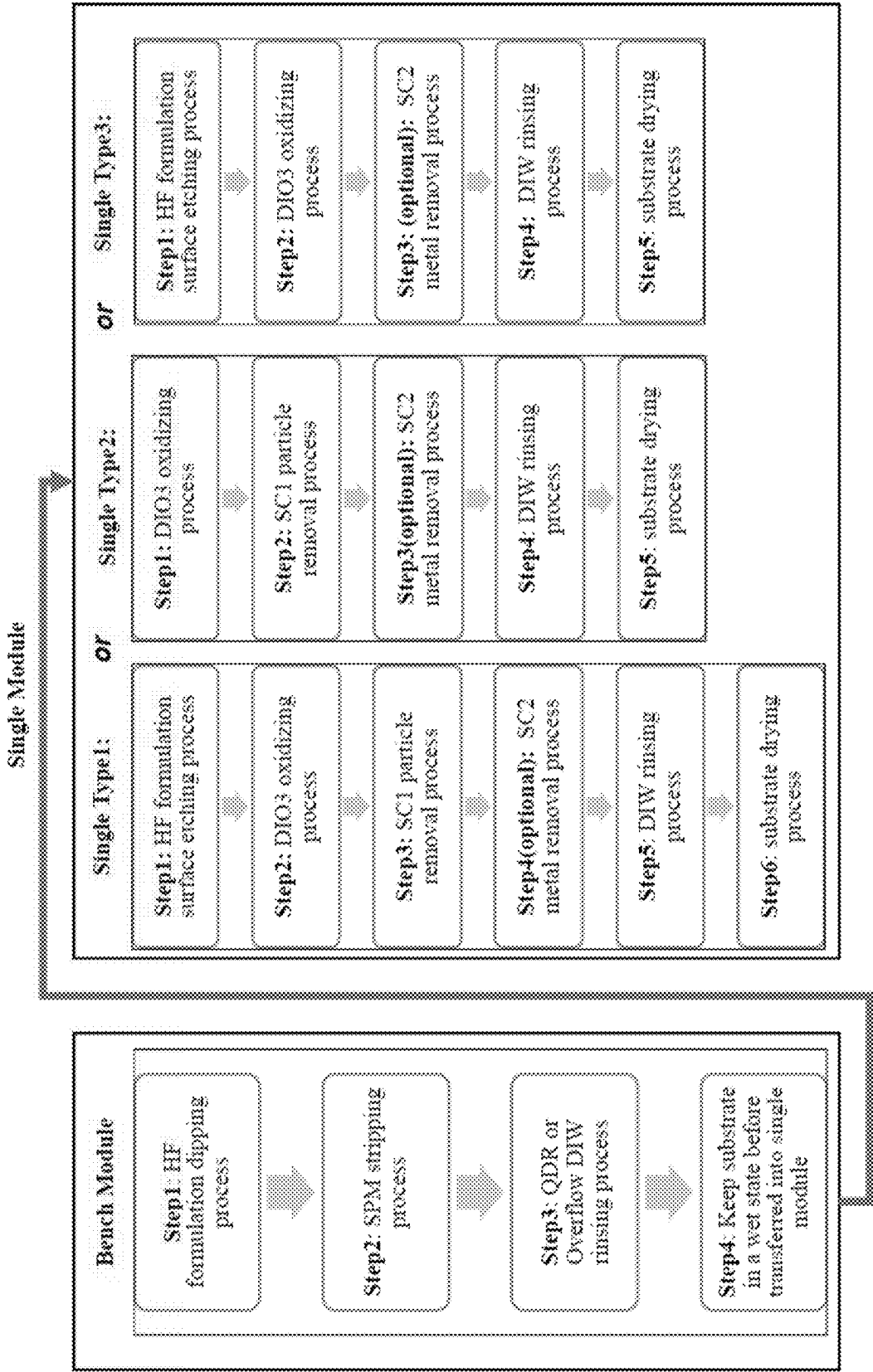


FIG 5

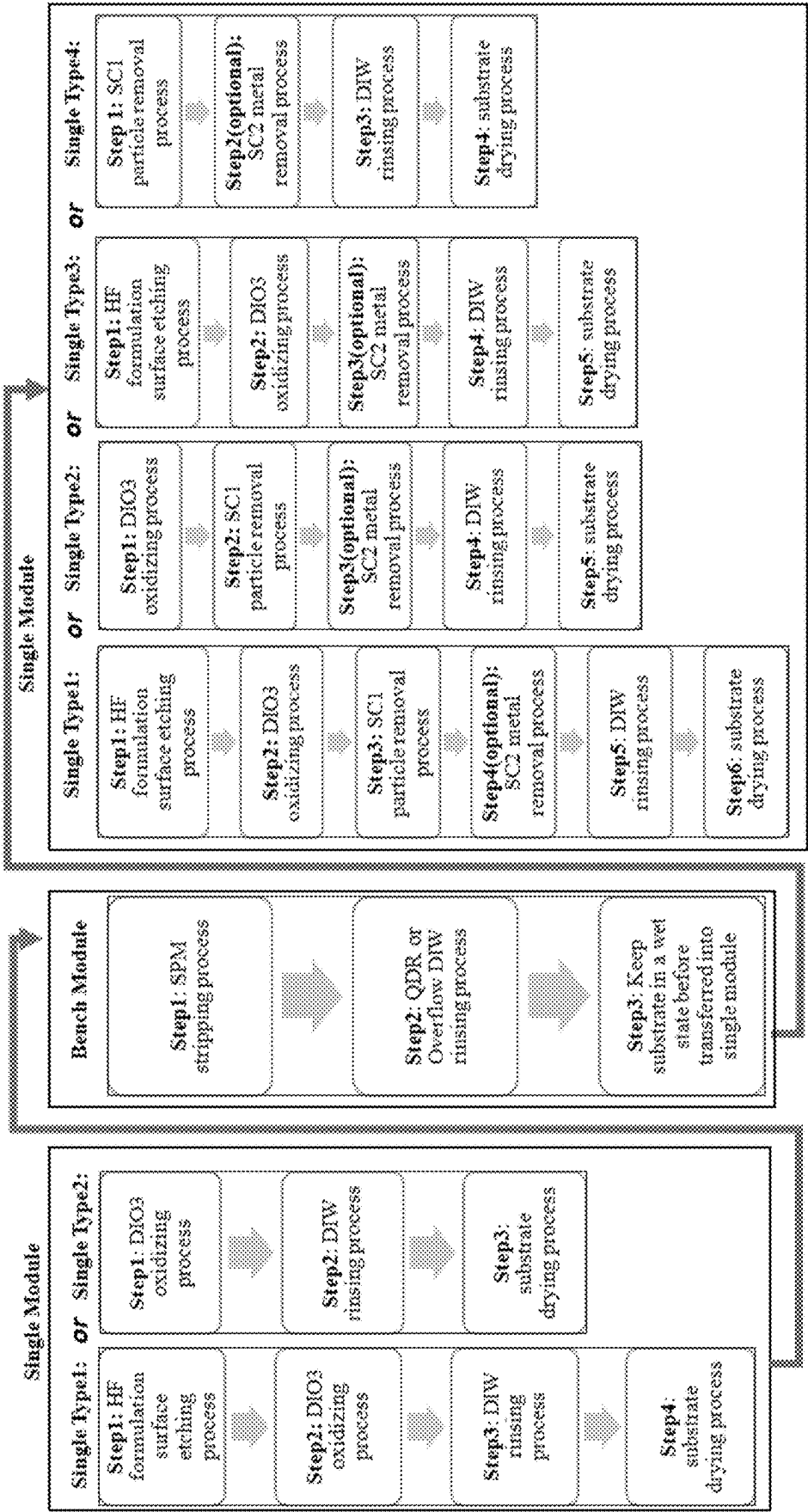


FIG 6

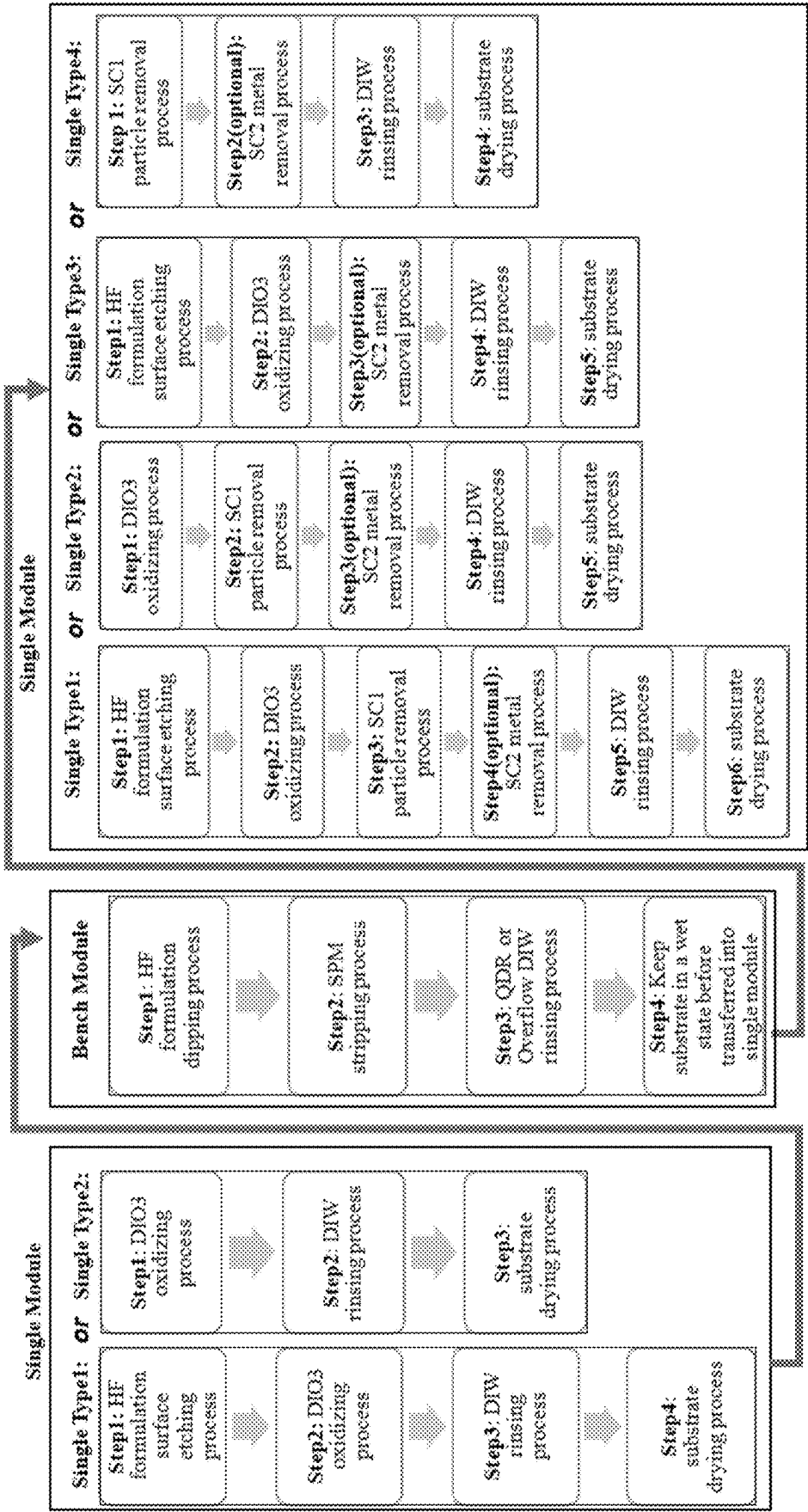


FIG 7



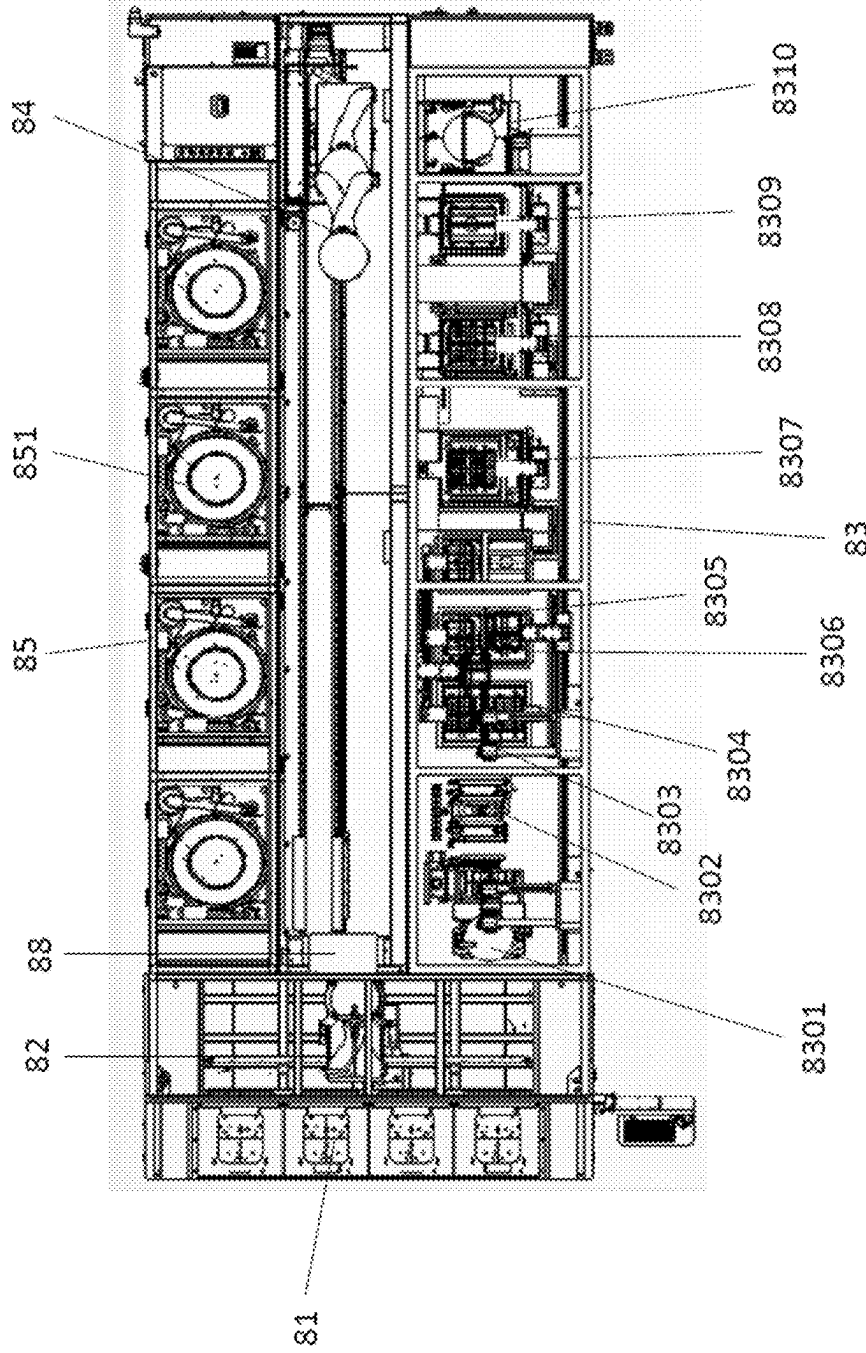


FIG 8

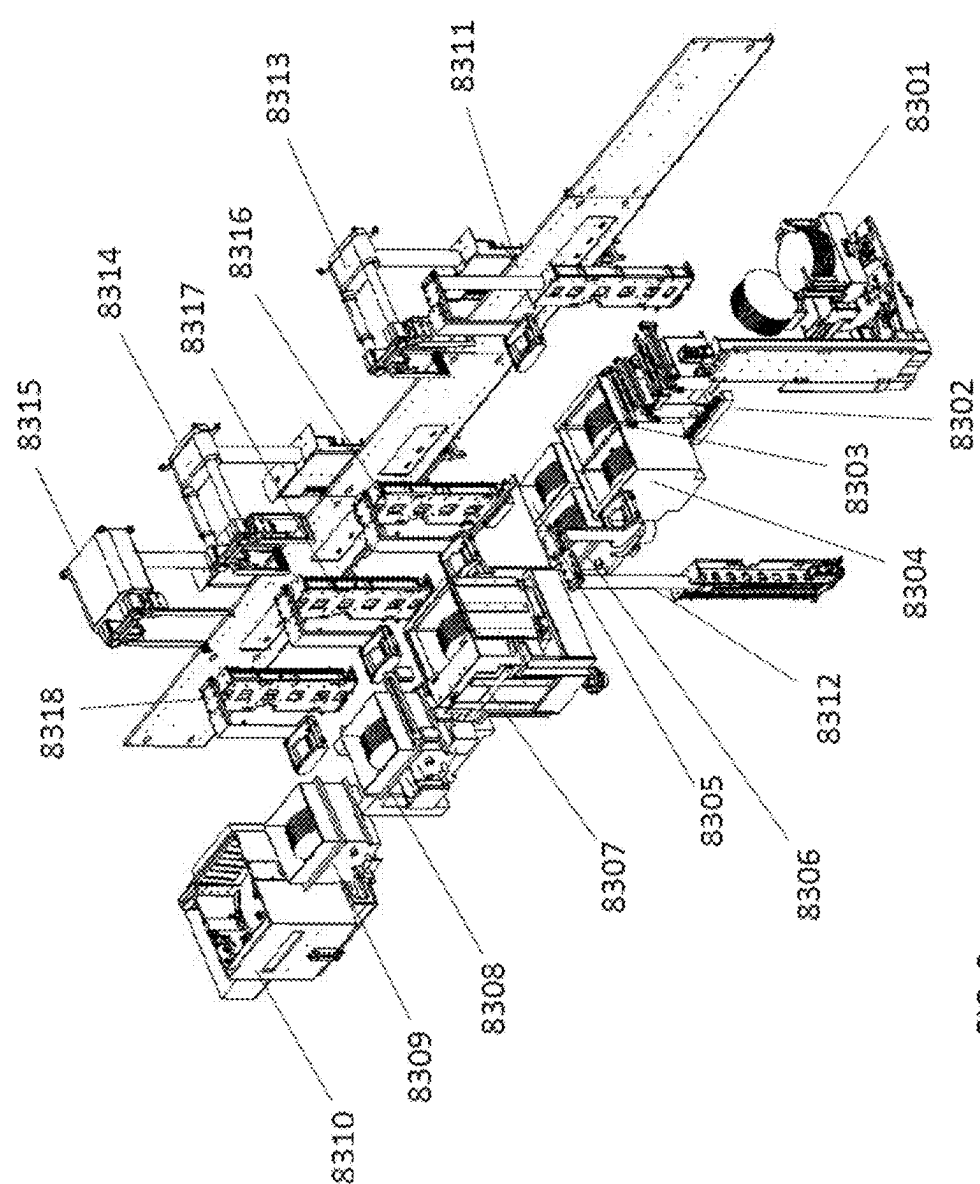


FIG 9

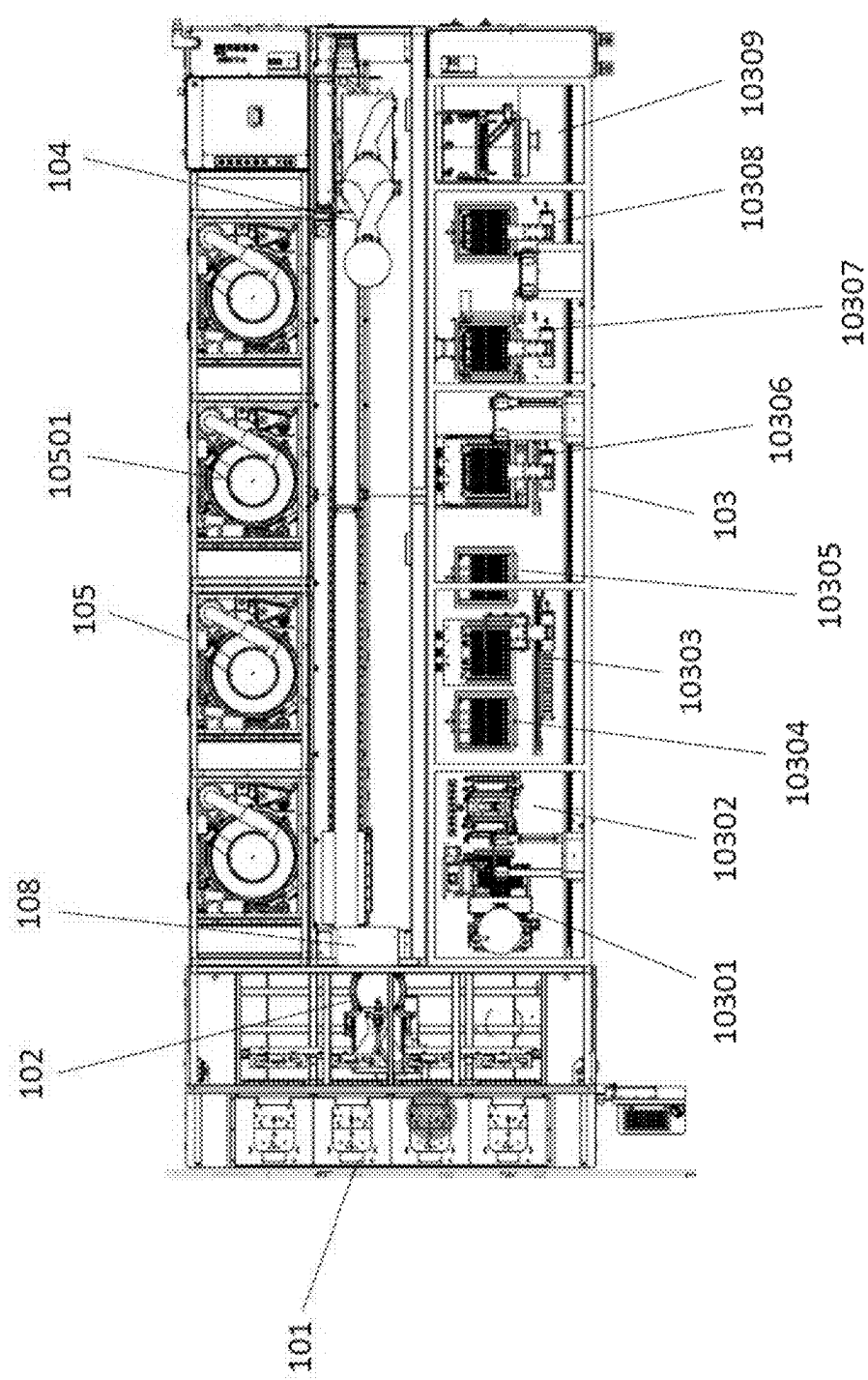


FIG 10

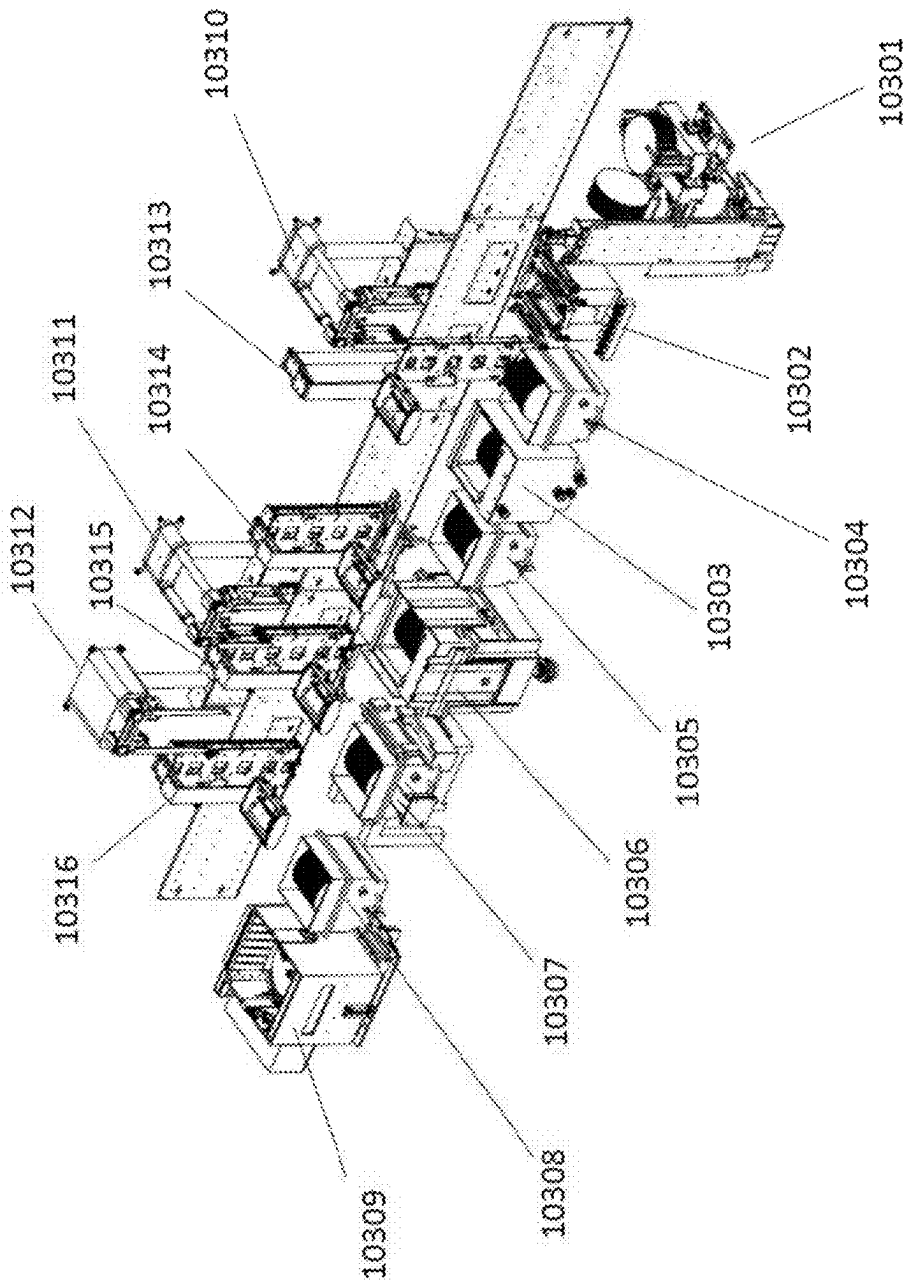


FIG 11

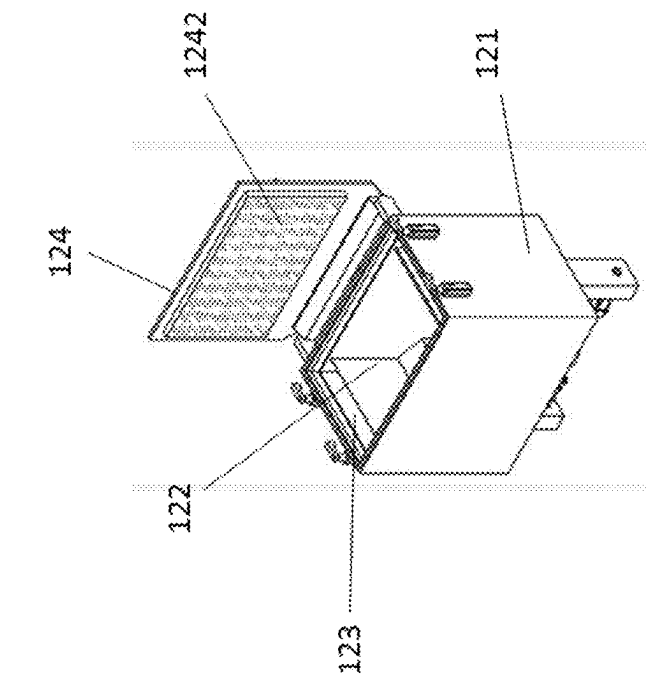


FIG 12B

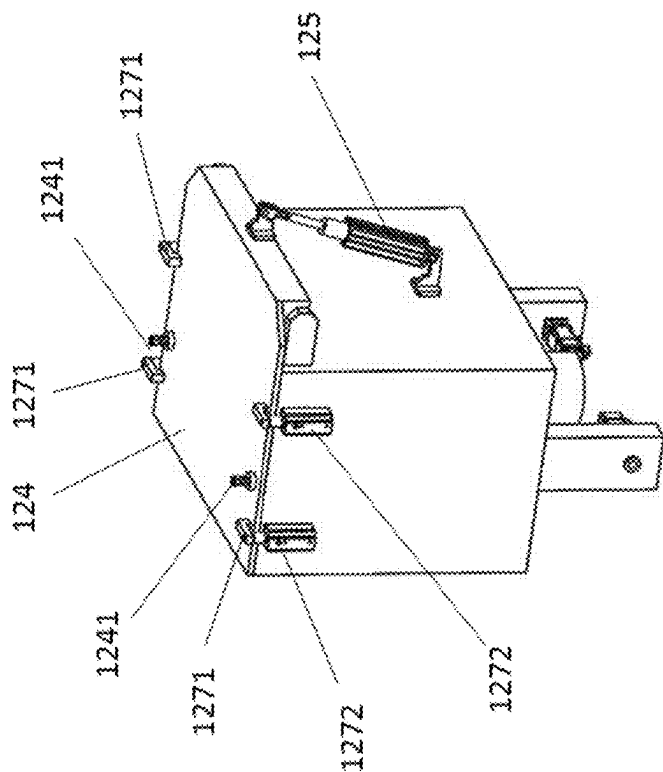


FIG 12A

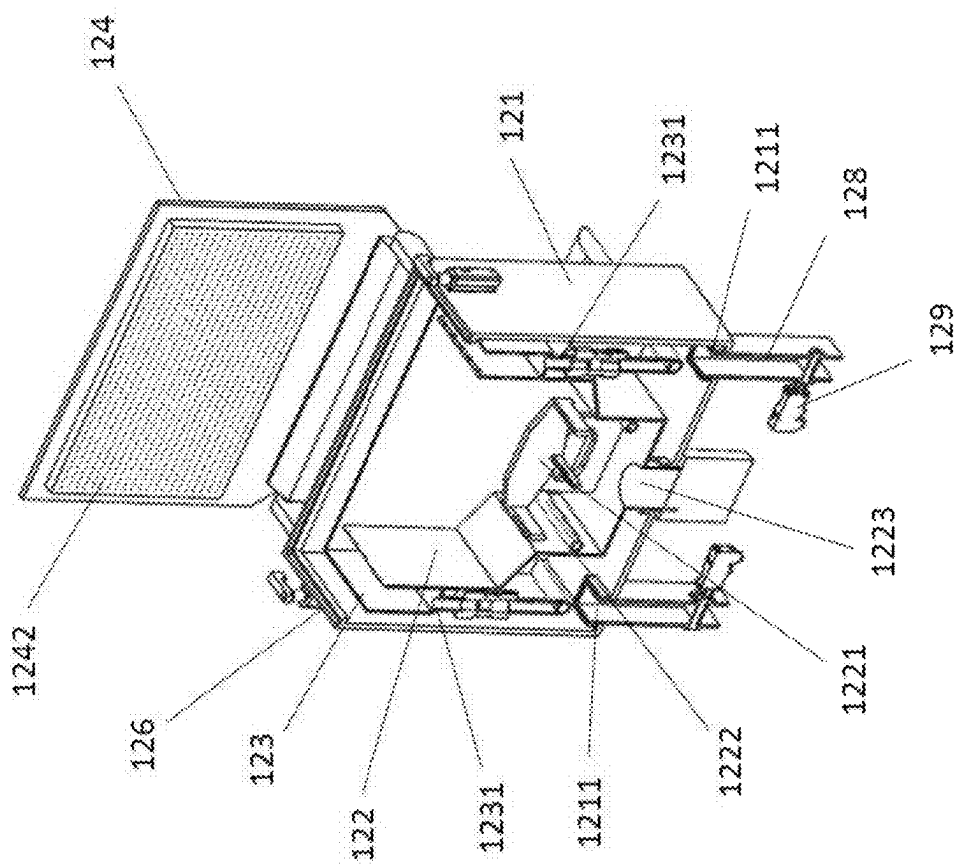


FIG 12C

## METHOD AND APPARATUS FOR REMOVING PARTICLES OR PHOTORESIST ON SUBSTRATES

### CROSS REFERENCE TO RELATED APPLICATIONS

**[0001]** The application is a divisional application of U.S. application Ser. No. 17/920,665, which is a national phase of international application No. PCT/CN2020/085948, filed on Apr. 21, 2020. The contents of these applications are incorporated herein by reference in their entirety.

### BACKGROUND OF THE INVENTION

#### 1. Field of the Invention

**[0002]** The present invention generally relates to a semiconductor wet cleaning field, and more particularly to a method and an apparatus for removing particles or photoresist on semiconductor substrates.

#### 2. The Related Art

**[0003]** Conventionally, organic photoresist strip processes were developed via using a combination of dry and wet treatments. However, dry treatments based on reactive plasma ashing have been shown to present issues, such as plasma induced damage, resist popping, incomplete resist removal, and byproduct redeposition, consequently, it requires follow up with a wet stripping/cleaning. To avoid plasma issues, wet stripping processes based on organic solvents and aggressive acid chemistries, such as aqueous mixtures of sulfuric acid and hydrogen peroxide (SPM), were developed. Nowadays, SPM is widely used for photoresist stripping and post-resist strip cleaning processes.

**[0004]** In the past tens of years, SPM wet stripping process is typically carried out in a wet bench tool because of its high throughput, while maintaining a low cost of operation. However, today's SPM bench-only wet process cannot achieve the cleaning performance required for advanced technology nodes due to (1) the particle removal efficiency by wet bench tool has been reduced significantly when the particle sizes are reduced to 45 nm and below; (2) not able to remove photoresists treated by high dose ( $>1E17$  ions/cm<sup>2</sup>) of ion implantation because of limited temperature of SPM bath with normal temperature of 145° C. Single substrate wet stripping process seems to be an alternative solution for new generation of IC manufacture node which needs to remove particles smaller than 45 nm and clean photoresist treated by high dose of ion implantation due to its higher temperature of SPM (180° C.). However, single substrate SPM process requires the SPM mixture to be heated to high temperatures with only a fraction of the hot SPM touching the substrate surface and most of the SPM spinning off from the substrate, which cannot be reused due to loss of chemical energy. This process results in a great consumption of sulfuric acid, what's worse, disposal of waste is expensive and harmful to the environment.

**[0005]** In order to reduce environmental impact caused by large amount of SPM consumption, utilizing ozone chemistry for photoresist removal to replace SPM wet stripping process has been tried and tested. However, the cleaning efficiency especially for photoresist treated by medium dose ( $1E15$  ions/cm<sup>2</sup>) and high dose ( $>1E17$  ions/cm<sup>2</sup>) of ion

implantation was much lower than that of high temperature ( $>180^{\circ}$  C.) of single substrate SPM process.

**[0006]** From a removal efficient and environmental point of view, an integrated cleaning system and process method that combines a traditional bench SPM cleaning module and a single substrate cleaning module into one wet-clean system is disclosed in PCT patent application No. PCT/CN2018/090227, filed on Jun. 7, 2018. This system overcomes the challenges of traditional bench-only and single-only wet cleaning systems, using a two-step approach to optimize the advantages of wet bench and single substrate cleaning, achieving a cleaning efficiency that is comparable to the single substrate SPM cleaning while, at the same time, it allows the SPM to be recycled and reused and it greatly reduces the consumption of sulfuric acid. However, SPM temperature in bath is critical for wet stripping process for photoresist treated by high dose of ion implantation. But, a higher temperature can cause rapid decomposition of H<sub>2</sub>O<sub>2</sub> or fast depletion of the H<sub>2</sub>O<sub>2</sub>, therefore SPM used in this system can only be heated as high as 150° C. It is clear that for the high energy-high dose implanted resist, 150° C. SPM cannot remove thick crust formed during ion implantation.

### SUMMARY

**[0007]** The present invention discloses a method and an apparatus to remove fine particles at high efficiency and to remove photoresist and residuals treated by high energy and high dose ion implantation with much lower consumption of SPM chemicals.

**[0008]** More specifically, the present invention discloses a combination of DIO<sub>3</sub> and SPM substrate photoresist wet stripping method and a substrate photoresist wet stripping apparatus.

**[0009]** According to one embodiment of the present invention, a method for removing particles or photoresist from substrates comprises the following procedures: processing one or more substrates in a SPM bath and then in a DI water rinsing bath in a bench module, then keeping the one or more substrates being in wet state and transferring the one or more substrates to subsequent one or more single chambers of a single module to perform single substrate DIO<sub>3</sub> cleaning for remaining residues removal and final cleaning.

**[0010]** According to another embodiment of the present invention, a method for removing particles or photoresist from substrates comprises the following procedures: processing one or more substrates in a DIO<sub>3</sub> bath then in a SPM bath and then in a DI water rinsing bath in a bench module, then keeping the one or more substrates being in wet state and transferring the one or more substrates to subsequent one or more single chambers of a single module to perform single substrate cleaning for remaining residues removal and final cleaning.

**[0011]** According to another embodiment of the present invention, a method for removing particles or photoresist from substrates comprises the following procedures: processing one or more substrates in a HF bath then in a DIO<sub>3</sub> bath then in a SPM bath and then in a DI water rinsing bath in a bench module, then keeping the one or more substrates being in wet state and transferring the one or more substrates to subsequent one or more single chambers of a single module to perform single substrate cleaning for remaining residues removal and final cleaning.

**[0012]** According to another embodiment of the present invention, a method for removing particles or photoresist

from substrates comprises the following procedures: processing one or more substrates in one or more single chambers of a single module to perform single substrate DIO<sub>3</sub> cleaning, then processing the one or more substrates in a SPM bath and then in a DI water rinsing bath in a bench module, then keeping the one or more substrates being in wet state and transferring the one or more substrates to the one or more single chambers of the single module to perform single substrate cleaning for remaining residues removal and final cleaning.

#### BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIG. 1 illustrates cleaning performance comparison between stand-alone bench SPM cleaning plus stand-alone single cleaning and a Bench-Single integrated cleaning.

[0014] FIG. 2 illustrates a bench DIO<sub>3</sub>-SPM combination process and a subsequent single substrate cleaning process in an integrated sequence according to an exemplary embodiment of the present invention.

[0015] FIG. 3 illustrates another bench DIO<sub>3</sub>-SPM combination process and a subsequent single substrate cleaning process in an integrated sequence according to another exemplary embodiment of the present invention.

[0016] FIG. 4 illustrates a bench SPM-based process and a subsequent single substrate cleaning process including a DIO<sub>3</sub> process in an integrated sequence according to an exemplary embodiment of the present invention.

[0017] FIG. 5 illustrates another bench SPM-based process and a subsequent single substrate cleaning process including a DIO<sub>3</sub> process in an integrated sequence according to another exemplary embodiment of the present invention.

[0018] FIG. 6 illustrates a single substrate DIO<sub>3</sub>-based cleaning process, a bench SPM-based process and a subsequent single substrate cleaning process in an integrated sequence according to an exemplary embodiment of the present invention.

[0019] FIG. 7 illustrates another single substrate DIO<sub>3</sub>-based cleaning process, a bench SPM-based process and a subsequent single substrate cleaning process in an integrated sequence according to another exemplary embodiment of the present invention.

[0020] FIG. 8 illustrates a top view of an apparatus for removing particles or photoresist on substrates according to an exemplary embodiment of the present invention.

[0021] FIG. 9 illustrates a perspective view of a bench module of the apparatus shown in FIG. 8.

[0022] FIG. 10 illustrates a top view of an apparatus for removing particles or photoresist on substrates according to another exemplary embodiment of the present invention.

[0023] FIG. 11 illustrates a perspective view of a bench module of the apparatus shown in FIG. 10.

[0024] FIG. 12A illustrates a perspective view of a DIO<sub>3</sub> bath according to an exemplary embodiment of the present invention.

[0025] FIG. 12B illustrates a perspective view of the DIO<sub>3</sub> bath of which shutter is open.

[0026] FIG. 12C illustrates a cross-sectional view of the DIO<sub>3</sub> bath shown in FIG. 12B.

#### DETAILED DESCRIPTION OF EMBODIMENTS

[0027] During the SPM cleaning process, the challenges experienced are that viscous SPM is hard to rinse off and hygroscopic sulfur (S) residues adhere to the substrate surface and absorb moisture, which causes particulate defects are hard to remove after drying. Therefore, after the bench SPM processing, the substrate has to be transferred into a single process chamber with a wet status, which prevents the substrate surface from drying out to form watermark defects or to absorb ionic and particulate pollutants from the external environment.

[0028] As shown in FIG. 1, a comparison test was designed to clarify cleaning performance between stand-alone bench SPM cleaning plus stand-alone single cleaning process and a Bench-Single integrated cleaning process.

Test 1: Stand-Alone Bench SPM Cleaning Plus Stand-Alone Single Cleaning

[0029] a) Sequence 1: bench process: SPM+QDR (quick dump rinse)+Dry;

[0030] More specifically, the bench process is performed in a bench tool. The bench process comprises the steps: processing at least one wafer with SPM, rinsing the wafer by a way of QDR, and at last drying the wafer.

[0031] b) Sequence 2: the wafer stays in a FOUP for 3 hours; and

[0032] c) Sequence 3: single process: SC1+N<sub>2</sub> Spray SC1+N<sub>2</sub> Dry.

[0033] More specifically, the single process is performed in a single wafer cleaning tool. The single process comprises the steps: processing the wafer with SC1, processing the wafer with SC1 which is atomized to generate fine droplets and accelerated by pressurized nitrogen gas through a gas-liquid atomizing spray nozzle (N<sub>2</sub> Spray SC1), rinsing the wafer with DIW, and at last drying the wafer with N<sub>2</sub>.

Test 2: Bench-Single Integrated Cleaning

[0034] Bench process of Bench-Single integrated cleaning: SPM+QDR, and

[0035] Single process of Bench-Single integrated cleaning: SC1+N<sub>2</sub> Spray SC1+N<sub>2</sub> Dry.

[0036] More specifically, the bench process and the single process are performed in a Bench-Single integrated cleaning tool. The Bench-Single integrated cleaning process comprises the steps: processing at least one wafer with SPM and then rinsing the wafer by a way of QDR in a bench module of the Bench-Single integrated tool, then transferring the wafer to a single module of the Bench-Single integrated tool to perform the single process which further comprises processing the wafer with SC1, processing the wafer with SC1 which is atomized to generate fine droplets and accelerated by pressurized nitrogen gas through a gas-liquid atomizing spray nozzle (N<sub>2</sub> Spray SC1), rinsing the wafer with DIW, and at least drying the wafer with N<sub>2</sub>.

[0037] Test 1 shows 40 nm particle add amount after the wafers have been run through the stand-alone bench SPM cleaning is 297~331. The wafers are then stored in the FOUP for 3 hours before being processed in the stand-alone single wafer cleaning tool, and 40 nm particle add amount shown in test 1 is decreased to 117~130. Also as shown in FIG. 1, the test 2 of Bench-Single integrated cleaning shows 40 nm particle add amount is decreased to ~1~9. FIG. 1 shows that the cleaning effect of the Bench-Single integrated



cleaning is significantly better than the stand-alone bench SPM cleaning plus the stand-alone single cleaning. The key conclusion for getting better defect removal performance is to keep the wafer surface in a wet state between the post bench SPM cleaning process and the pre single wafer cleaning process. Because the defects are very difficult to remove even by a single wafer cleaning process after the wafer has been dried.

**[0038]** During the SPM cleaning process, it is very critical to control and keep a certain thickness of liquid film on the wafer surface between the post bench SPM cleaning process and the pre single wafer cleaning process, which can prevent the wafer surface from drying out to form watermark defects or to absorb ionic and particulate pollutants from the external environment. However, it should be noted that in actual application, even with a Bench-Single integrated cleaning system, a perfect wafer wetting status is hard to achieve, so that a DIO3 and SPM combination cleaning process is disclosed in the present invention to eliminate the impacts during the wafer being transferred from the bench module to the single module. Furthermore, the residua of sulfur by products and particles attached on the wafer surface after the SPM process is harder to remove by SC1 only process.

**[0039]** In order to clarify a DIO3 enhanced SPM cleaning performance, tests are carried out in a Bench-Single integrated cleaning tool with 12 inch bare silicon wafer, and particle amounts are measured on a KLA-Tencor Surfscan SP5 with 19 nm metrology recipe. As shown in Table 1, the test conditions are below:

**[0040]** Test 1: Bench process in the Bench-Single integrated cleaning tool: SPM+Hot QDR, plus Single process in the Bench-Single integrated cleaning tool: SC1+N2 Spray SC1+N2 Dry.

**[0041]** Test 2: Bench process in the Bench-Single integrated cleaning tool: SPM+Hot QDR, plus Single process in the Bench-Single integrated cleaning tool: O3+SC1+N2 Spray SC1+N2 Dry.

**[0042]** Test 3: Bench process in the Bench-Single integrated cleaning tool: SPM+Hot QDR, plus Single process in the Bench-Single integrated cleaning tool: DHF+O3+SC1+N2 Spray SC1+N2 Dry.

cleaning process shows a positive cleaning effect. Here, HF will undercut particles attached on the wafer surface, and DIO3 will further remove sulfur by products and conditioning the wafer surface after the HF process.

**[0044]** For photoresist irradiated or treated by high dose ( $>1E17$  ion/cm<sup>2</sup>) and high energy ( $>10$  KeV) ion implantation, the C—H inside photoresist are disordered, cross-linked and formed double bonds C=C or graphite structure. Those graphite structure materials form hard crust on the surface of photoresist, and are hard to remove unless using very high temperature ( $>180^{\circ}$  C.) SPM with single wafer cleaning tool. However, with wet bench process, the temperature of SPM in a bath can only be heated to  $150^{\circ}$  C. or less due to quick decomposition of H<sub>2</sub>O<sub>2</sub>. Therefore it is critical to remove the crust which is being hardened in the ion implantation.

**[0045]** O<sub>3</sub> is a very strong oxidant. The DIO<sub>3</sub> process directly oxidizes C=C double bonds, and the resist becomes thinner during this reaction. Taking the ozonolysis of alkenes mechanism for example, it involves the attack of ozone on the C=C double bonds and forms the molozonide intermediate which is quite unstable. Due to this unstable nature, the molozonide continues reacting—breaking apart to form a carbonyl molecule and a carbonyl oxide molecule. The carbonyl molecule and the carbonyl oxide molecule formed in the first step rearrange themselves, reforming to create a more stable ozonide intermediate. This ozonide intermediate can be subjected to either an oxidative workup or a reductive workup. The oxidative workup will give carboxylic acid as the product whereas the reductive workup will yield aldehydes or ketones.

**[0046]** After the crust being treated by DIO<sub>3</sub>, and the remaining by products will be removed by SPM with temperature less than  $150^{\circ}$  C.

**[0047]** Therefore, for removing particles or photoresist, especially for removing photoresist treated by medium dose or high dose of ion implantation, the present invention discloses a plurality of methods which combine DIO<sub>3</sub> and SPM in one cleaning sequence.

**[0048]** According to one embodiment of the present invention, a method for removing particles or photoresist on

TABLE 1

| DIO <sub>3</sub> enhanced SPM cleaning tests in a Bench-Single integrated cleaning tool |               |                                       |                           |      |                   |      |                           |                |          |        |                                   |
|-----------------------------------------------------------------------------------------|---------------|---------------------------------------|---------------------------|------|-------------------|------|---------------------------|----------------|----------|--------|-----------------------------------|
| Item                                                                                    | Bench process | Single process                        | PRE <sup>②</sup>          |      | POST <sup>②</sup> |      | I <sup>②</sup> (Post-Pre) |                |          |        | PRE Total                         |
|                                                                                         |               |                                       | 19 um                     | 1 um | 19 um             | 1 um | 19-30 um                  | 30-40 um       | 40-60 um | >60 um |                                   |
| Bench-Single integrated cleaning process                                                | SPM-Hot QDR   | Test 1: SC1 <sup>②</sup> 2 Spray      | 521                       | 0    | 528               | 0    | -1 <sup>②</sup>           | 7              | 3        | -2     | - <sup>②</sup>                    |
|                                                                                         |               | SC1 <sup>②</sup> 2 Dry                | 301                       | 0    | 3 <sup>②</sup>    | 0    | -8                        | 11             | 9        | -7     | - <sup>②</sup> %                  |
|                                                                                         |               | Test 2: O3-SC1 <sup>②</sup> Spray     | 281                       | 0    | <sup>②</sup>      | 0    | -62                       | - <sup>②</sup> | -7       | -2     | -7 <sup>②</sup> 28%               |
|                                                                                         |               | SC1 <sup>②</sup> Dry                  | <sup>②</sup> <sup>②</sup> | 0    | 198               | 0    | -55                       | -5             | 1        | -15    | -7 <sup>②</sup> 27%               |
|                                                                                         |               | Test 3: DHF-O3-SC1 <sup>②</sup> Spray | 261                       | 0    | 191               | 0    | -56                       | -3             | -8       | -3     | -70 37%                           |
|                                                                                         |               | SC1 <sup>②</sup> Dry                  | 283                       | 0    | 19 <sup>②</sup>   | 0    | -64                       | -8             | -3       | -3     | - <sup>②</sup> 7 3 <sup>②</sup> % |

<sup>②</sup> indicates text missing or illegible when filed

**[0043]** Test shows combine bench SPM- hot QDR cleaning and the single cleaning in one sequence, after the single cleaning process with SC1-N2 Spray SC1, particle has a little added. The single cleaning process with O3-SC1-N2 Spray SC1 can enhance particle remove. The single cleaning process with DHF-O3-SC1-N2 Spray SC1 has the best particle remove efficiency. The DIO3 and SPM combination

substrates comprises a bench DIO3-SPM combination process and a subsequent single cleaning process in an integrated sequence, the method comprising:

**[0049]** transferring one or more substrates into a DIO3 solution accommodated in a DIO3 bath;

**[0050]** after the one or more substrates are processed in the DIO3 bath, taking the one or more substrates out

from the DIO3 bath and transferring the one or more substrates into a SPM solution accommodated in a SPM bath;

[0051] after the one or more substrates are processed in the SPM bath, taking the one or more substrates out from the SPM bath and rinsing the one or more substrates; and

[0052] transferring the one or more substrates to one or more single chambers to perform single substrate cleaning and drying process.

[0053] More specifically, referring to FIG. 2, the bench DIO3-SPM combination process is performed in a bench module of a Bench-Single integrated cleaning apparatus. The bench DIO3-SPM combination process comprises:

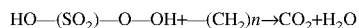
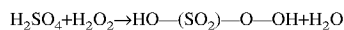
[0054] transferring one or more substrates into a DIO3 solution accommodated in a DIO3 bath to perform a DIO<sub>3</sub> oxidizing process, wherein the DIO<sub>3</sub> is used here for photoresist pre-treatment, including oxidizing C=C double bonds and softening the photoresist crust, which would facilitate the subsequent photoresist stripping process;

[0055] after the one or more substrates are processed in the DIO3 bath, taking the one or more substrates out from the DIO3 bath and transferring the one or more substrates into a SPM solution accommodated in a SPM bath to perform a SPM stripping process, wherein the SPM is used here for removing the softened photoresist crust and the bulk resist by the intense chemical reaction; and

[0056] after the one or more substrates are processed in the SPM bath, taking the one or more substrates out from the SPM bath and transferring the one or more substrates to a DIW bath to perform a rinsing process.

[0057] The sulfuric mixtures act as aggressive strippers during SPM cleaning process. Two types of stripping mechanisms are widely accepted. The first is dehydration mechanism: the first action is that sulfuric acid can cause organic carbonization and dehydration, by which the resist is firstly undercut and then floats away of the substrate. The second is oxidation mechanism: the mixing of sulfuric acid and hydrogen peroxide forms Caro's acid (H<sub>2</sub>SO<sub>5</sub>), which oxidizes the carbonized resist product into CO and CO<sub>2</sub> as an extremely strong oxidant.

[0058] The reactions in the SPM cleaning process are:



[0059] The SPM temperature in the SPM bath is critical for removing photoresist treated by high dose of ion implantation. However, a higher temperature can cause rapid decomposition of H<sub>2</sub>O<sub>2</sub> or fast depletion of the H<sub>2</sub>O<sub>2</sub>. Therefore, the SPM used in the bench module can only be heated as high as 150° C. It is clear that for the high energy-high dose implanted resist, 150° C. SPM cannot remove the thick photoresist crust formed during ion implantation.

[0060] Therefore, the present invention discloses a method by adding DIO<sub>3</sub> (ozonized DIW) process prior to the SPM (<150° C.) process. O<sub>3</sub> is a very strong oxidant and can directly oxidize C=C double bonds, so that the thick crust of resist formed during high energy and high dose ion implantation can be softened during this reaction. After the crust of resist being softened, the underneath resist can be removed by SPM with temperature less than 150° C.

[0061] In an embodiment, the SPM solution is a mixture of H<sub>2</sub>SO<sub>4</sub> and H<sub>2</sub>O<sub>2</sub>, and H<sub>2</sub>SO<sub>4</sub> and H<sub>2</sub>O<sub>2</sub> mix ratio is 3:1 to 50:1, the temperature of the mixture is 80° C. to 150° C.

[0062] In an embodiment, the rinsing process in the DIW bath comprises QDR (Quick dump drain) and overflow rinse.

[0063] In an embodiment, the ozone concentration of DIO3 solution in the DIO3 bath is 30 ppm to 120 ppm.

[0064] In an embodiment, the flow rate of DIO3 solution supplied to the DIO3 bath is 10LPM to 30LPM.

[0065] In an embodiment, before the one or more substrates are transferred to the DIO3 bath, the liquid in the DIO3 bath is DIW. A procedure of the DIO3 oxidizing process further comprises:

[0066] step 1: opening a shutter of the DIO3 bath, transferring the one or more substrates into the DIW in the DIO3 bath, closing the shutter of the DIO3 bath, in order to prevent ozone gas leakage, before the one or more substrates are transferred to the DIO3 bath, the liquid in the DIO3 bath is DIW;

[0067] step 2: overflowing an ozonized DIW from the DIO3 bath bottom to replace the DIW in the DIO3 bath;

[0068] step 3: after the DIO3 bath is full of the ozonized DIW, keeping the ozonized DIW overflowing for a calculated time, in an embodiment, the time is 5 to 15 min, better 10 min;

[0069] step 4: quick dump drain the ozonized DIW;

[0070] step 5: filling the DIO3 bath with pure DIW;

[0071] step 6: opening the shutter of the DIO3 bath, taking the one or more semiconductor substrates out from the DIO3 bath.

[0072] In the step 3, in consideration of compensating for half-life of ozone destruction and maintaining a constant ozone concentration, the DIO3 supply flow rate must be high enough to refill the DIO3 bath, and make sure the ozone concentration would not drop too much during process. As well known, the ozonized DIW decays rapidly and the half-life of ozonized DIW at 25° C. and pH 7.0 is typically about 15 minutes. How to compensate for half-life of ozone destruction and maintain a constant ozone concentration is important. In order to assure its strong oxidizing property, the decreasing of DIO3 concentration during process should be controlled less than D=10 ppm, and the process should have a full consideration of ozone decomposition rate, fresh ozonized DIW replenishing flow rate, and an optimal DIO3 process time.

[0073] Assume t is the half-life of DIO3, C is the target concentration of DIO3, V is the volume of the DIO3 bath, the average ozone decomposition rate can be estimated as

$$d = \frac{1}{2} * \frac{C}{t}$$

(instant ozone decomposition rate will vary and be bigger at higher concentration and smaller at lower concentration), the fresh ozonized DIW replenishing flow rate is  $r = V/(D/d)$ . The DIO3 process time is in the range of 5 min to 15 min, better to be 10 min.

[0074] In an embodiment, the ozone concentration of DIO3 solution in the DIO3 bath is 90 ppm, the half-life of DIO3 is 15 min, the volume of the DIO3 bath is 50 L, the average ozone decomposition rate is  $d = 0.5 * 90 / 15 = 3$  ppm/min, the fresh ozonized DIW replenishing flow rate is

$r=50/(10/3)=15$  LPM. In consideration of using average ozone decomposition rate in the calculation, it is better to use  $2r=30$  LPM in practical application.

**[0075]** In the present invention, the DIO3 bath combines both ozonized DIW and pure DIW in one bath, and the DIO3 process and DIW rinsing process can be performed in one bath, solving the environment problem caused by ozone gas escaped from DIO3 liquid which smells pungent and is a toxic gas which may cause harmful effects on the health.

**[0076]** After the one or more substrates are performed the rinsing process in the DIW bath and before the one or more substrates are transferred to the one or more single chambers of a single module of the Bench-Single integrated cleaning apparatus, keep the one or more substrates being in wet state. More specifically, transferring the one or more substrates to a wetting buffer area to keep the substrates being in wet status.

**[0077]** Please continue to refer to FIG. 2. In one embodiment of the present invention, the single substrate cleaning and drying process is performed in one single chamber of a single module of the Bench-Single integrated cleaning apparatus. The single substrate cleaning and drying process in one single chamber comprises:

**[0078]** spraying HF formulation solution on the substrate surface to perform a surface etching process;

**[0079]** spraying DIO3 on the substrate surface to perform a DIO3 oxidizing process;

**[0080]** spraying SC1 on the substrate surface to perform a particle removal process;

**[0081]** spraying DIW on the substrate surface to perform a DIW rinsing process; and

**[0082]** drying the substrate.

**[0083]** Optionally, after the particle removal process by using the SC1, the single substrate cleaning and drying process can further comprise spraying SC2 on the substrate surface to perform metal removal process.

**[0084]** In another embodiment of the present invention, the single substrate cleaning and drying process in one single chamber comprises:

**[0085]** spraying DIO3 on the substrate surface to perform a DIO3 oxidizing process;

**[0086]** spraying SC1 on the substrate surface to perform a particle removal process;

**[0087]** spraying DIW on the substrate surface to perform a DIW rinsing process; and

**[0088]** drying the substrate.

**[0089]** Optionally, after the particle removal process by using the SC1, the single substrate cleaning and drying process can further comprise spraying SC2 on the substrate surface to perform metal removal process.

**[0090]** In yet another embodiment of the present invention, the single substrate cleaning and drying process in one single chamber comprises:

**[0091]** spraying HF formulation solution on the substrate surface to perform a surface etching process;

**[0092]** spraying DIO3 on the substrate surface to perform a DIO3 oxidizing process;

**[0093]** spraying DIW on the substrate surface to perform a DIW rinsing process; and

**[0094]** drying the substrate.

**[0095]** Optionally, after the DIO3 oxidizing process, the single substrate cleaning and drying process can further comprise spraying SC2 on the substrate surface to perform metal removal process.

**[0096]** In yet another embodiment of the present invention, the single substrate cleaning and drying process in one single chamber comprises:

**[0097]** spraying SC1 on the substrate surface to perform a particle removal process;

**[0098]** spraying DIW on the substrate surface to perform a DIW rinsing process; and

**[0099]** drying the substrate.

**[0100]** Optionally, after the particle removal process by using the SC1, the single substrate cleaning and drying process can further comprise spraying SC2 on the substrate surface to perform metal removal process.

**[0101]** Referring to FIG. 3, according to another embodiment of the present invention, a method for removing particles or photoresist on substrates comprises a bench DIO3-SPM combination process and a subsequent single cleaning process in an integrated sequence. Comparing with the method disclosed in FIG. 2, the difference is that the method disclosed in FIG. 3 further comprises: transferring one or more substrates into a HF formulation solution accommodated in a HF formulation bath to perform a HF formulation dipping process before the one or more substrates are transferred to the DIO3 bath.

**[0102]** In an embodiment, the HF formulation solution is a mixture of HF and DIW, and HF and DIW mix ratio is 100:1 to 1000:1.

**[0103]** In another embodiment, the HF formulation solution is a BOE mixture. HF weight percentage of the BOE mixture is 0.05% to 10%, and  $\text{NH}_4\text{F}$  weight percentage of the BOE mixture is 10% to 40%.

**[0104]** According to one embodiment of the present invention, a method for removing particles or photoresist on substrates comprises a bench SPM-based process and a subsequent DIO3-based single cleaning process in an integrated sequence, the method comprising:

**[0105]** transferring one or more substrates into a SPM solution accommodated in a SPM bath;

**[0106]** after the one or more substrates are processed in the SPM bath, taking the one or more substrates out from the SPM bath and rinsing the one or more substrates; and

**[0107]** transferring the one or more substrates to one or more single chambers to perform single substrate cleaning and drying process, wherein the single substrate cleaning and drying process comprises at least one DIO3 oxidizing process.

**[0108]** The purpose of using DIO3 is to remove residual of Sulfur by products after SPM process and residuals of photoresist which have not been removed by SPM process in the previous bench module. The ozone concentration of DIO3 is 30 ppm to 120 ppm, and 80 ppm is preferred.

**[0109]** More specifically, referring to FIG. 4, the bench SPM-based process is performed in a bench module of a Bench-Single integrated cleaning apparatus. The bench SPM-based process comprises:

**[0110]** transferring one or more substrates into a SPM solution accommodated in a SPM bath to perform a SPM stripping process, wherein the SPM is used here for removing photoresist crust and the bulk resist by the intense chemical reaction with concentrated sulfuric acid in the bench module; and

**[0111]** after the one or more substrates are processed in the SPM bath, taking the one or more substrates out

from the SPM bath and transferring the one or more substrates to a DIW bath to perform a rinsing process.

[0112] In an embodiment, the SPM solution is a mixture of  $\text{H}_2\text{SO}_4$  and  $\text{H}_2\text{O}_2$ , and  $\text{H}_2\text{SO}_4$  and  $\text{H}_2\text{O}_2$  mix ratio is 3:1 to 50:1, the temperature of the mixture is 80° C. to 150° C.

[0113] In an embodiment, the rinsing process in the DIW bath comprises QDR (Quick dump drain) and overflow rinse.

[0114] After the one or more substrates are performed the rinsing process in the DIW bath and before the one or more substrates are transferred to the one or more single chambers of a single module of the Bench-Single integrated cleaning apparatus, keep the one or more substrates being in wet state. More specifically, transferring the one or more substrates to a wetting buffer area to keep the substrates being in wet status.

[0115] Please continue to refer to FIG. 4. In one embodiment of the present invention, the DIO3-based single substrate cleaning and drying process is performed in one single chamber of a single module of the Bench-Single integrated cleaning apparatus. The DIO3-based single substrate cleaning and drying process in one single chamber comprises:

[0116] spraying HF formulation solution on the substrate surface to perform a surface etching process;

[0117] spraying DIO3 on the substrate surface to perform a DIO3 oxidizing process;

[0118] spraying SC1 on the substrate surface to perform a particle removal process;

[0119] spraying DIW on the substrate surface to perform a DIW rinsing process; and

[0120] drying the substrate.

[0121] Optionally, after the particle removal process by using the SC1, the single substrate cleaning and drying process can further comprise spraying SC2 on the substrate surface to perform metal removal process.

[0122] In another embodiment of the present invention, the DIO3-based single substrate cleaning and drying process is performed in one single chamber of a single module of the Bench-Single integrated cleaning apparatus. The DIO3-based single substrate cleaning and drying process in one single chamber comprises:

[0123] spraying DIO3 on the substrate surface to perform a DIO3 oxidizing process;

[0124] spraying SC1 on the substrate surface to perform a particle removal process;

[0125] spraying DIW on the substrate surface to perform a DIW rinsing process; and

[0126] drying the substrate.

[0127] Optionally, after the particle removal process by using the SC1, the single substrate cleaning and drying process can further comprise spraying SC2 on the substrate surface to perform metal removal process.

[0128] In yet another embodiment of the present invention, the DIO3-based single substrate cleaning and drying process is performed in one single chamber of a single module of the Bench-Single integrated cleaning apparatus. The DIO3-based single substrate cleaning and drying process in one single chamber comprises:

[0129] spraying HF formulation solution on the substrate surface to perform a surface etching process;

[0130] spraying DIO3 on the substrate surface to perform a DIO3 oxidizing process;

[0131] spraying DIW on the substrate surface to perform a DIW rinsing process; and

[0132] drying the substrate.

[0133] Optionally, after the DIO3 oxidizing process, the single substrate cleaning and drying process can further comprise spraying SC2 on the substrate surface to perform metal removal process.

[0134] Referring to FIG. 5, according to another embodiment of the present invention, a method for removing particles or photoresist on substrates comprises a bench SPM-based process and a subsequent DIO3-based single cleaning process in an integrated sequence. Comparing with the method disclosed in FIG. 4, the difference is that the method disclosed in FIG. 5 further comprises: transferring one or more substrates into a HF formulation solution accommodated in a HF formulation bath to perform a HF formulation dipping process before the one or more substrates are transferred to the SPM bath. Specifically, transferring the one or more substrates into a HF formulation solution accommodated in a HF formulation bath, after the one or more substrates are processed in the HF formulation bath, taking the one or more substrates out from the HF formulation bath and rinsing the one or more substrates, and then transferring the one or more substrates to the SPM bath. The HF formulation solution can attack photoresist, and remove Si-containing polymers and fluorocarbon polymers from the substrate sidewall, and would be benefit for subsequent SPM photoresist removal.

[0135] In an embodiment, the HF formulation solution is a mixture of HF and DIW, and HF and DIW mix ratio is 100:1 to 1000:1.

[0136] In another embodiment, the HF formulation solution is a BOE mixture. HF weight percentage of the BOE mixture is 0.05% to 10%, and  $\text{NH}_4\text{F}$  weight percentage of the BOE mixture is 10% to 40%.

[0137] According to one embodiment of the present invention, a method for removing particles or photoresist on substrates comprises a DIO3-based single substrate cleaning and drying process for photoresist attacking and thinning, a subsequent bench SPM-based process, and a final single substrate cleaning and drying process in an integrated sequence, the method comprising:

[0138] transferring one or more substrates to one or more single chambers to perform single substrate cleaning and drying process, wherein the single substrate cleaning and drying process comprises at least one DIO3 oxidizing process;

[0139] after the one or more substrates are processed in the one or more single chambers, transferring the one or more substrates into a SPM solution accommodated in a SPM bath;

[0140] after the one or more substrates are processed in the SPM bath, taking the one or more substrates out from the SPM bath and rinsing the one or more substrates; and

[0141] transferring the one or more substrates to the one or more single chambers to perform single substrate cleaning and drying process.

[0142] More specifically, referring to FIG. 6, in an embodiment, the DIO3-based single substrate cleaning and drying process prior to the bench SPM-based process is performed in one single chamber of a single module of a Bench-Single integrated cleaning apparatus. The DIO3-based single substrate cleaning and drying process com-

prises: a HF formulation surface etching process, a DIO3 oxidizing process, a DIW rinsing process and a drying process.

**[0143]** In another embodiment, the DIO3-based single substrate cleaning and drying process prior to the bench SPM-based process is performed in one single chamber of a single module of a Bench-Single integrated cleaning apparatus. The DIO3-based single substrate cleaning and drying process comprises: a DIO3 oxidizing process, a DIW rinsing process and a drying process.

**[0144]** In an embodiment, the bench SPM-based process is performed in a bench module of the Bench-Single integrated cleaning apparatus. The bench SPM-based process comprises: a SPM stripping process and a DIW rinsing process.

**[0145]** After the one or more substrates are performed the DIW rinsing process in the bench module and before the one or more substrates are transferred to the one or more single chambers, keep the one or more substrates being in wet state. More specifically, transferring the one or more substrates to a wetting buffer area to keep the substrates being in wet status.

**[0146]** Similar to the method disclosed in FIG. 2, in one embodiment, the single substrate cleaning and drying process after the bench SPM-based process is performed in one single chamber of the single module of the Bench-Single integrated cleaning apparatus. The single substrate cleaning and drying process comprises: a HF formulation surface etching process, a DIO3 oxidizing process, a SC1 particle removal process, a DIW rinsing process and a drying process. Optionally, after the SC1 particle removal process, the single substrate cleaning and drying process can further comprise a SC2 metal removal process.

**[0147]** In another embodiment, the single substrate cleaning and drying process after the bench SPM-based process is performed in one single chamber of the single module of the Bench-Single integrated cleaning apparatus. The single substrate cleaning and drying process comprises: a DIO3 oxidizing process, a SC1 particle removal process, a DIW rinsing process and a drying process. Optionally, after the SC1 particle removal process, the single substrate cleaning and drying process can further comprise a SC2 metal removal process.

**[0148]** In yet another embodiment, the single substrate cleaning and drying process after the bench SPM-based process is performed in one single chamber of the single module of the Bench-Single integrated cleaning apparatus. The single substrate cleaning and drying process comprises: a HF formulation surface etching process, a DIO3 oxidizing process, a DIW rinsing process and a drying process. Optionally, after the DIO3 oxidizing process, the single substrate cleaning and drying process can further comprise a SC2 metal removal process.

**[0149]** In yet another embodiment, the single substrate cleaning and drying process after the bench SPM-based process is performed in one single chamber of the single module of the Bench-Single integrated cleaning apparatus. The single substrate cleaning and drying process comprises: a SC1 particle removal process, a DIW rinsing process and a drying process. Optionally, after the SC1 particle removal process, the single substrate cleaning and drying process can further comprise a SC2 metal removal process.

**[0150]** Referring to FIG. 7, according to another embodiment of the present invention, a method for removing particles or photoresist on substrates comprises a DIO3-

based single substrate cleaning and drying process for photoresist attacking and thinning, a subsequent bench SPM-based process, and a final single substrate cleaning and drying process in an integrated sequence. Comparing with the method disclosed in FIG. 6, the difference is that the method disclosed in FIG. 7 further comprises: a HF formulation dipping process prior to the SPM stripping process.

**[0151]** In all embodiments described above in the present invention, the DIO<sub>3</sub> is solution dissolved ozone in DIW. The SC1 is a mixture of ammonium hydroxide and hydrogen peroxide in deionized water, sometime called “standard clean-1”. The SC2 is a mixture of hydrochloric acid and hydrogen peroxide in deionized water, sometime called “standard clean-2”.

**[0152]** It should be recognized that in the single chamber, one DIW rinsing process can be inserted between any two chemical processes. For example, between the DIO3 oxidizing process and the SC1 particle removal process, one DIW rinsing process can be inserted there between.

**[0153]** The present invention discloses an apparatus for removing particles or photoresist on substrates. The apparatus includes a bench module configured to implement one or more substrates bench cleaning process, a single module having multiple single chambers configured to implement single substrate cleaning and drying process and a process robot configured to transfer the one or more substrates between the bench module and the single module, wherein the bench module has at least one DIO3 bath configured to accommodate DIO3 solution for processing the one or more substrates, at least one SPM bath configured to accommodate SPM solution for processing the one or more substrates, at least one DIW bath configured to accommodate DIW for rinsing the one or more substrates, and at least one second substrate transfer robot configured to transfer the one or more substrates among the DIO3 bath, the SPM bath and the DIW bath.

**[0154]** More specifically, referring to FIG. 8 and FIG. 9, an apparatus for removing particles or photoresist on substrates according to one exemplary embodiment of the present invention is illustrated. The apparatus includes a plurality of, e.g. four load ports **81** each of which receives a FOUN, an index robot **82**, a buffer **88**, a bench module **83**, a process robot **84** and a single module **85** having multiple single chambers **851**.

**[0155]** The bench module **83** includes a substrate loader **8301**, a cleaning bath **8302**, a first HF formulation bath **8303**, a second HF formulation bath **8304**, a first DIO3 bath **8305**, a second DIO3 bath **8306**, a SPM bath **8307**, a first DIW bath **8308**, a second DIW bath **8309**, a wetting buffer area **8310**, a first substrate transfer robot **8313**, a second substrate transfer robot **8314**, a third substrate transfer robot **8315**, a first lifter **8311**, a second lifter **8312**, a third lifter **8316**, a fourth lifter **8317** and a fifth lifter **8318**.

**[0156]** When the apparatus is used for removing particles or photoresist on substrates, the process sequence is as follow.

**[0157]** The index robot **82** gets one or more substrates from one FOUN and transfers the one or more substrates to the substrate loader **8301** by one time or several times. The substrate loader **8301** holds the one or more substrates and then makes the one or more substrates rotate 90 degrees along the horizontal axis and rotate another 90 degrees along the vertical axis so that the one or more substrates are vertically held by the substrate loader **8301**.

[0158] The first substrate transfer robot **8313** gets the one or more substrates, e.g. 13 pieces of substrates or 12 pieces of substrates from the substrate loader **8301** and transfers the one or more substrates to the first lifter **8311**. The first lifter **8311** holds the one or more substrates and brings the one or more substrates to immerse into HF formulation solution accommodated in the first HF formulation bath **8303**. The first substrate transfer robot **8313** gets another one or more substrates from the substrate loader **8301** and transfers the one or more substrates to the second lifter **8312**. The second lifter **8312** holds the one or more substrates and brings the one or more substrates to immerse into HF formulation solution accommodated in the second HF formulation bath **8304**.

[0159] After the process in the first HF formulation bath **8303** or the process in the second HF formulation bath **8304** completes, the second substrate transfer robot **8314** gets the one or more substrates from the first HF formulation bath **8303** and transfers the one or more substrates to the first DIO3 bath **8305**, or the second substrate transfer robot **8314** gets the one or more substrates from the second HF formulation bath **8304** and transfers the one or more substrates to the second DIO3 bath **8306**. In order to reduce the risk of water mark formed on the substrate surface, it is better to control the transfer time to be less than 20 seconds, and better to be less than 10 seconds. The first lifter **8311** and the second lifter **8312** respectively rise to be above the first HF formulation bath **8303** and the second HF formulation bath **8304**. In this embodiment, the first HF formulation bath **8303**, the second HF formulation bath **8304**, the first DIO3 bath **8305**, and the second DIO3 bath **8306** are arranged in two rows, and each row has one HF formulation bath and one adjacent DIO3 bath.

[0160] After the process in the first DIO3 bath **8305** or the process in the second DIO3 bath **8306** completes, the second substrate transfer robot **8314** gets the one or more substrates from the first DIO3 bath **8305** or the second DIO3 bath **8306** and transfers the one or more substrates to the third lifter **8316**. The third lifter **8316** brings the one or more substrates to immerse into SPM solution accommodated in the SPM bath **8307**.

[0161] After the process in the SPM bath **8307** completes, the third lifter **8316** rises to be above the SPM bath **8307** and the second substrate transfer robot **8314** gets the one or more substrates from the third lifter **8316** and transfers the one or more substrates to the fourth lifter **8317** or the fifth lifter **8318**. The fourth lifter **8317** or the fifth lifter **8318** brings the one or more substrates to the first DIW bath **8308** or the second DIW bath **8309** to implement QDR (Quick dump drain) or overflow rinsing process.

[0162] After the one or more substrates are processed in the first DIW bath **8308** or the second DIW bath **8309**, the fourth lifter **8317** or the fifth lifter **8318** rises to be above the first DIW bath **8308** or the second DIW bath **8309**, and the third substrate transfer robot **8315** gets the one or more substrates from the fourth lifter **8317** or the fifth lifter **8318** and transfers the one or more substrates to the wetting buffer area **8310** by one time or several times. In the wetting buffer area **8310**, the one or more substrates can be rotated from vertical to horizontal and kept in wet status before the one or more substrates are transferred to the one or more single chambers **851** of the single module **85**. During this waiting period, the substrates in the wetting buffer area **8310** can be sprayed by DI water to keep the substrate surface wet.

[0163] The process robot **84** gets the one or more substrates from the wetting buffer area **8310** and transfers the one or more substrates to the one or more single chambers **851** to implement single substrate cleaning and drying process.

[0164] After the single substrate cleaning and drying process complete in one single chamber **851**, the process robot **84** gets the substrate from the single chamber **851** and transfers it to the buffer **88**, and then the index robot **82** gets the substrate from the buffer **88** and transfers it back to the FOUN.

[0165] In the bench module **83**, the cleaning bath **8302** is used for cleaning the second substrate transfer robot **8314** while the second substrate transfer robot **8314** is idle.

[0166] Referring to FIG. 10 and FIG. 11, an apparatus for removing particles or photoresist on substrates according to another exemplary embodiment of the present invention is illustrated. The apparatus includes a plurality of, e.g. four load ports **101** each of which receives a FOUN, an index robot **102**, a buffer **108**, a bench module **103**, a process robot **104** and a single module **105** having multiple single chambers **10501**.

[0167] The bench module **103** includes a substrate loader **10301**, a cleaning bath **10302**, a HF formulation bath **10303**, a first DIO3 bath **10304**, a second DIO3 bath **10305**, a SPM bath **10306**, a first DIW bath **10307**, a second DIW bath **10308**, a wetting buffer area **10309**, a first substrate transfer robot **10310**, a second substrate transfer robot **10311**, a third substrate transfer robot **10312**, a first lifter **10313**, a third lifter **10314**, a fourth lifter **10315** and a fifth lifter **10316**.

[0168] Comparing with the apparatus shown in FIG. 8 and FIG. 9, the apparatus shown in this embodiment has one HF formulation bath **10303** and the HF formulation bath **10303** is set between the first DIO3 bath **10304** and the second DIO3 bath **10305**. Comparing with the apparatus shown in FIG. 8 and FIG. 9, since the apparatus shown in this embodiment is lack of a second HF formulation bath, and accordingly, the apparatus shown in this embodiment is lack of a second lifter. The substrate transfer sequence is similar to that of embodiment shown in FIG. 8 and FIG. 9, which will not be repeat described herein.

[0169] Please refer to FIGS. 12A-12C. A DIO3 bath according to an exemplary embodiment of the present invention is illustrated. The DIO3 bath has an outer bath **121**, an inner bath **122**, two overflow grooves **123**, a shutter **124** and a driving device **125**.

[0170] The bottom of the outer bath **121** sets two exhaust ports **1211**. Two exhaust lines **128** are respectively connected to the two exhaust ports **1211** of the outer bath **121**. A pressure monitor and a pressure damper **129** are respectively set on the two exhaust lines **128**. The material of the outer bath **121** can select PVC.

[0171] The inner bath **122** is set in the outer bath **121** and configured to accommodate DIO3 solution. The inner bath **122** has a substrate holding pedestal **1221** configured to supporting and holding the one or more substrates. Two liquid inlet pipes **1222** are set in the inner bath **122** and configured to supply liquid to the inner bath **122**. A drain port **1223** is set at the bottom of the inner bath **122** and configured to drain the liquid in the inner bath **122** to the outside of the DIO3 bath. Preferably, the drain port **1223** is a quick drain port. The inner bath **122** is made of high purity quartz materials.

[0172] The two overflow grooves 123 are set at both sides of the inner bath 122. Each overflow groove 123 has two drain holes 1231 configured to drain the liquid in the overflow groove 123 to the outside of the DIO3 bath.

[0173] The shutter 124 is set on the outer bath 121 to seal the outer bath 121. The shutter 124 has at least one gas intake 1241 set on the top side of the shutter 124 and a plurality of exhaust holes 1242 set on the bottom side of the shutter 124. The driving device 125 is connected to the shutter 124 and configured to open or close the shutter 124.

[0174] For better seal, preferably, a seal ring 126 is set on the top of the outer bath 121. At least one pair of locking devices are set on the outer bath 121 to lock the shutter 124 while the shutter 124 is closed. Each locking device has a lock head 1271 and an actuator 1272 connected to the lock head 1271 and driving the lock head 1271 to rotate and rise and fall. After the shutter 124 is closed, the actuator 1272 drives the lock head 1271 to rotate to be above the shutter 124 and then the actuator 1272 drives the lock head 1271 to fall so that the lock head 1271 locks the shutter 124. When the shutter 124 needs to open, the actuator 1272 drives the lock head 1271 to rise and then the actuator 1272 drives the lock head 1271 to rotate to make the lock head 1271 leave the shutter 124 so that the shutter 124 can be open.

[0175] During process, when the one or more substrates are transferred into the DIO3 bath and held by the substrate holding pedestal 1221, the shutter 124 is closed to seal the DIO3 bath to prevent ozone gas from leaking to the peripheral environment. Ozone smells pungent and it is a toxic gas which may cause harmful effects on the health. In that case, for safety protection, how to seal the DIO3 bath and keep a very low ambient ozone level is challenging and important.

[0176] In this embodiment, the inner bath 122 is used for accommodating DIO3. The two overflow grooves 123 are used as a DIO3 overflow area. The outer bath 121 is used for exhausting gas. There is a gap between the top of the inner bath 122 and the top of the outer bath 121, which can ensure enough area for exhausting gas, preventing the ozone gas from leaking from the outer bath 121. Furthermore, the present invention also discloses an ozone gas anti-leakage structure. Specifically, a purge gas, such as N2 or CDA is supplied to the outer bath 121 through the at least one gas intake 1241 set on the top side of the shutter 124 and a plurality of exhaust holes 1242 set on the bottom side of the shutter 124. Purge gas is provided into the outer bath 121, serving as a gas carrier to prevent the ozone gas from leaking to the peripheral environment. Meanwhile, it's important to control the exhaust pressure in the outer bath 121 to balance the gas intake and output, which can be realized by the pressure monitor and the pressure damper 129.

[0177] The foregoing description of the present invention has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise form disclosed, and obviously many modifications and variations are possible in light of the above teaching. Such modifications and variations that may be apparent to those skilled in the art are intended to be included within the scope of this invention as defined by the accompanying claims.

What is claimed is:

1. An apparatus for removing particles or photoresist on substrates, comprising:

a bench module, configured to implement one or more substrates bench cleaning process;

a single module, having multiple single chambers configured to implement single substrate cleaning and drying process; and

a process robot, configured to transfer the one or more substrates between the bench module and the single module;

wherein the bench module has at least one DIO<sub>3</sub> bath configured to accommodate DIO<sub>3</sub> solution for processing the one or more substrates, at least one SPM bath configured to accommodate SPM solution for processing the one or more substrates, at least one DIW bath configured to accommodate DIW for rinsing the one or more substrates, and at least one second substrate transfer robot configured to transfer the one or more substrates among the DIO<sub>3</sub> bath, the SPM bath and the DIW bath.

2. The apparatus as claimed in claim 1, wherein the bench module further comprises at least one HF formulation bath configured to accommodate HF formulation solution for processing the one or more substrates.

3. The apparatus as claimed in claim 2, wherein the bench module has two DIO<sub>3</sub> baths and two HF formulation baths, the two DIO<sub>3</sub> baths and the two HF formulation baths are arranged in two rows, and each row has one HF formulation bath and one adjacent DIO<sub>3</sub> bath.

4. The apparatus as claimed in claim 3, wherein the bench module further comprises:

a substrate loader, configured to hold and rotate the one or more substrates;

a first lifter and a second lifter, the first lifter and the second lifter being corresponding to the two HF formulation baths and configured to hold the one or more substrates; and

a first substrate transfer robot;

wherein the first substrate transfer robot gets the one or more substrates from the substrate loader and transfers the one or more substrates to the first lifter or the second lifter, the first lifter or the second lifter brings the one or more substrates to immerse into the HF formulation solution accommodated in the corresponding HF formulation bath, after the one or more substrates are processed in the HF formulation bath, the second substrate transfer robot gets the one or more substrates from the HF formulation bath and transfers the one or more substrates to the adjacent DIO<sub>3</sub> bath, the first lifter or the second lifter rises to be above the two HF formulation baths.

5. The apparatus as claimed in claim 2, wherein the bench module has two DIO<sub>3</sub> baths and one HF formulation bath, the HF formulation bath is arranged between the two DIO<sub>3</sub> baths.

6. The apparatus as claimed in claim 1, wherein the bench module has a wetting buffer area to keep the one or more substrates being in wet status before the one or more substrates are transferred to the one or more single chambers.

7. The apparatus as claimed in claim 1, wherein the bench module has a cleaning bath configured to clean the second substrate transfer robot.

8. The apparatus as claimed in claim 1, wherein the DIO<sub>3</sub> bath comprises:

an outer bath, the bottom of the outer bath sets at least one exhaust port;

an inner bath, the inner bath is set in the outer bath and configured to accommodate  $\text{DIO}_3$  solution, the inner bath has a substrate holding pedestal configured to supporting and holding the one or more substrates, at least one liquid inlet pipe set in the inner bath and configured to supply liquid to the inner bath, and a drain port set at the bottom of the inner bath and configured to drain the liquid in the inner bath to the outside of the  $\text{DIO}_3$  bath;

two overflow grooves, the two overflow grooves are set at both sides of the inner bath, each overflow groove has at least one drain hole configured to drain the liquid in the overflow groove to the outside of the  $\text{DIO}_3$  bath;

a shutter, the shutter is set on the outer bath to seal the outer bath, the shutter has at least one gas intake set on the top side of the shutter and a plurality of exhaust holes on the bottom side of the shutter; and

a driving device, the driving device is connected to the shutter and configured to open or close the shutter.

**9.** The apparatus as claimed in claim **8**, further comprising a seal ring set on the top of the outer bath, at least one pair of locking devices are set on the outer bath to lock the shutter while the shutter is closed.

**10.** The apparatus as claimed in claim **9**, wherein each locking device has a lock head and an actuator connected to the lock head and driving the lock head to rotate and rise and fall.

**11.** The apparatus as claimed in claim **8**, further comprising at least one exhaust line connected to the at least one exhaust port of the outer bath, at least one pressure monitor and at least one pressure damper are respectively set on the at least one exhaust line.

**12.** The apparatus as claimed in claim **1**, wherein a driving device is configured to open a shutter of the  $\text{DIO}_3$  bath, the second substrate transfer robot is configured to transfer the one or more substrates into a DIW in the  $\text{DIO}_3$  bath, the driving device is configured to close the shutter of the  $\text{DIO}_3$  bath, and then the  $\text{DIO}_3$  bath is configured to perform the following steps:

overflow a  $\text{DIO}_3$  solution from a bottom of the  $\text{DIO}_3$  bath

to replace the DIW in the  $\text{DIO}_3$  bath;

quick dump drain the  $\text{DIO}_3$  solution;

fill the  $\text{DIO}_3$  bath with pure DIW; and

then the driving device is configured to open the shutter of the  $\text{DIO}_3$  bath, the second substrate transfer robot is configured to take the one or more substrates out from the  $\text{DIO}_3$  bath.

**13.** The apparatus as claimed in claim **12**, wherein after the  $\text{DIO}_3$  bath is full of the  $\text{DIO}_3$  solution, the  $\text{DIO}_3$  bath is configured to keep the  $\text{DIO}_3$  solution overflowing for 5 to 15 min.

**14.** The apparatus as claimed in claim **12**, wherein ozone concentration of the  $\text{DIO}_3$  solution is 30 ppm to 120 ppm.

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